

MBM27C512-20-X

CMOS 512K-BIT UV EPROM

CMOS 524,288 BIT UV ERASABLE AND ELECTRICALLY PROGRAMMABLE READ ONLY MEMORY

The Fujitsu MBM27C512 is a high speed 524,288 bit static CMOS erasable and electrically reprogrammable read only memory (EPROM). It is especially well suited for application where rapid turn-around and/or bit pattern experimentation, and low-power consumption are important.

A 28-pin dual-in line package with a transparent lid and 32-Pad Leadless Chip Carrier (LCC) are used to package the MBM27C512. The transparent lid allows the user to expose the device to ultraviolet light in order to erase the memory bit pattern previously programmed. At the completion of erasure, a new pattern can then be written into the memory.

The MBM27C512 is fabricated using CMOS double polysilicon gate technology with single transistor stacked gate cells. It is organized as 65,536 words by 8 bits for use in microprocessor applications. Single +5V operation greatly facilitates its use in systems.

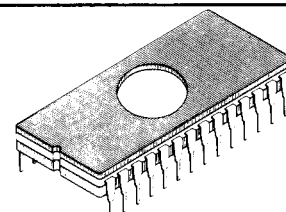
This specification is applied to "HW"-version.

- CMOS power consumption
Standby: 550 μ W max.
Active: 165mW max.
- 65,536 words x 8 bits organization, fully decoded
- Single location programming
- Programmable utilizing the Quick Programming Algorithm
- No clocks required (fully static operation)
- TTL compatible inputs/outputs
- Fast access time:
200ns max. (MBM27C512-20-X)
- Three-state output with OR-tie capability
- Output Enable ($\overline{OE}$) pin for simplified memory expansion
- Single +5V supply, $\pm 10\%$ tolerance
- Standard 28-pin Ceramic DIP: (Suffix: Z)
- Standard 32-pad Ceramic LCC: (Suffix: TV)

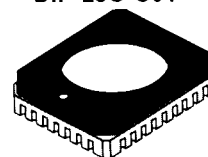
ABSOLUTE MAXIMUM RATINGS (see NOTE)

Rating	Symbol	Value	Unit
Temperature under Bias	TBIAS	-50 to +95	$^{\circ}\text{C}$
Storage Temperature	TSTG	-65 to +125	$^{\circ}\text{C}$
All Inputs/Outputs Voltage with respect to GND	VIN, VOUT	-0.6 to VCC +0.3	V
Voltage on A9 with respect to GND	VA9	-0.6 to +13.5	V
VPP Voltage with respect to GND	VPP	-0.6 to +14	V
Supply Voltage with respect to GND	VCC	-0.6 to +7	V

NOTE: Permanent device damage may occur if the above **Absolute Maximum Ratings** are exceeded. Functional operation should be restricted to the conditions as detailed in the operational sections of this data sheet. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

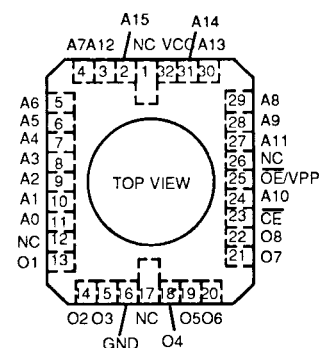
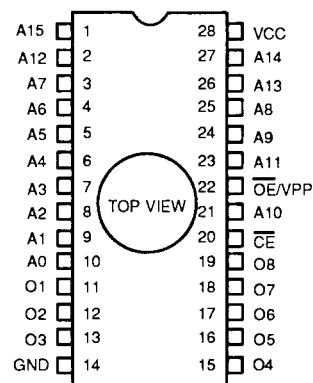


CERAMIC PACKAGE
DIP-28C-C01



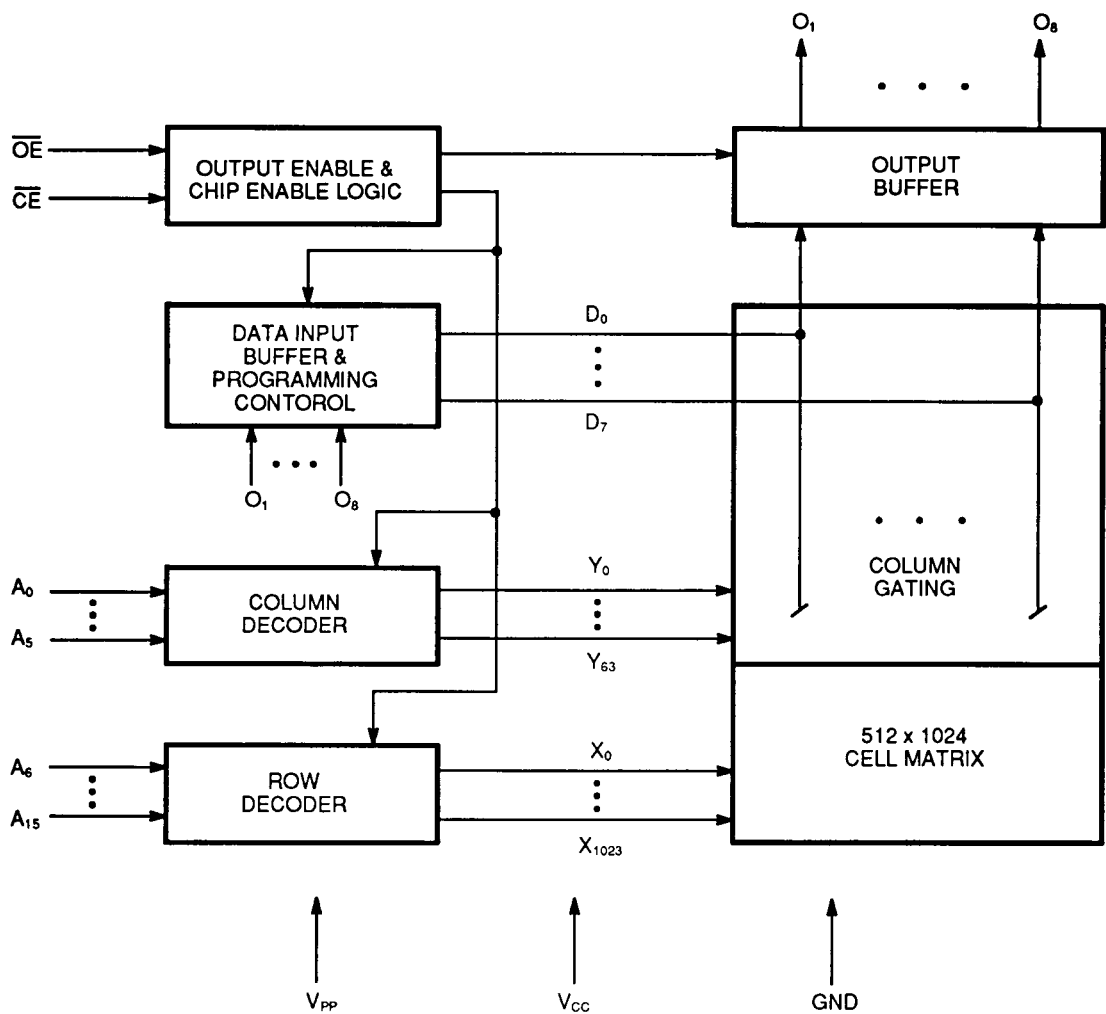
CERAMIC PACKAGE
LCC-32C-F01

PIN ASSIGNMENT



This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields. However, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high impedance circuit.

Fig. 1 – MBM27C512 BLOCK DIAGRAM



CAPACITANCE (T_A = 25°C, f = 1MHz)

Parameter	Symbol	Value			Unit
		Min	Typ	Max	
Input Capacitance (V _{IN} = 0V, except $\overline{OE}/V_{PP}$)	C _{IN1}	—	4	6	pF
$\overline{OE}/V_{PP}$ Input Capacitance (V _{IN} = 0V)	C _{IN2}	—	—	20	pF
Output Capacitance (V _{OUT} = 0V)	C _{OUT}	—	8	12	pF

FUNCTIONS AND PIN CONNECTIONS

Mode \ Function	Address Input	Data I/O	$\overline{CE}$	$\overline{OE}/V_{PP}$	V_{CC}	GND
Read	A_{IN}	D_{OUT}	V_{IL}	V_{IL}	5V	GND
Output Disable	A_{IN}	High-Z	V_{IL}	V_{IH}	5V	GND
Standby	Don't Care	High-Z	V_{IH}	Don't Care	5V	GND
Program	A_{IN}	D_{IN}	V_{IL}	12.5V	6V	GND
Program Verify	A_{IN}	D_{OUT}	V_{IL}	V_{IL}	6V	GND
Program Inhibit	Don't Care	High-Z	V_{IH}	12.5V	6V	GND

RECOMMENDED OPERATING CONDITIONS

(Referenced to GND)

Parameter	Symbol	Value			Unit
		Min	Typ	Max	
V_{CC} Supply Voltage	V_{CC}	4.5	5.0	5.5	V
Operating Temperature	T_A	-40		+85	°C

DC CHARACTERISTICS

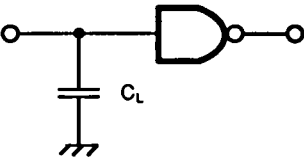
(Recommended operating conditions unless otherwise noted)

Parameter	Symbol	Value			Unit
		Min	Typ	Max	
Input Leakage Current ($V_{IN} = 5.5V$)	$ I_{LI} $			10	μA
Output Leakage Current ($V_{OUT} = 5.5V$)	$ I_{LO} $			10	μA
V_{CC} Standby Current ($\overline{CE} = V_{IH}$)	I_{SB1}			1	mA
V_{CC} Standby Current ($\overline{CE} = V_{CC} \pm 0.3V$, $I_{OUT} = 0mA$)	I_{SB2}		1	100	μA
V_{CC} Active Current ($\overline{CE} = V_{IL}$, $I_{OUT} = 0mA$)	I_{CC1}		4	30	mA
V_{CC} Operation Current ($f = 4MHz$, $I_{OUT} = 0mA$)	I_{CC2}		10	30	mA
Input High Voltage	V_{IH}	2.0		$V_{CC} + 0.3$	V
Input Low Voltage	V_{IL}	-0.1		0.8	V
Output Low Voltage ($I_{OL} = 2.1mA$)	V_{OL}			0.45	V
Output High Voltage ($I_{OH} = -400\mu A$)	V_{OH1}	2.4			V
Output High Voltage ($I_{OH} = -100\mu A$)	V_{OH2}	$V_{CC} - 0.7$			V

MBM27C512-20-X

Fig. 2 – AC TEST CONDITIONS (INCLUDING PROGRAMMING)

Input Pulse Levels: 0.45V to 2.4V
Input Rise/Fall Times: $\leq 20\text{ns}$
Timing Measurement Reference Levels: 0.8V and 2.0V for inputs
0.8V and 2.0V for outputs
Output Load: 1 TTL gate and $C_L = 100\text{pF}$



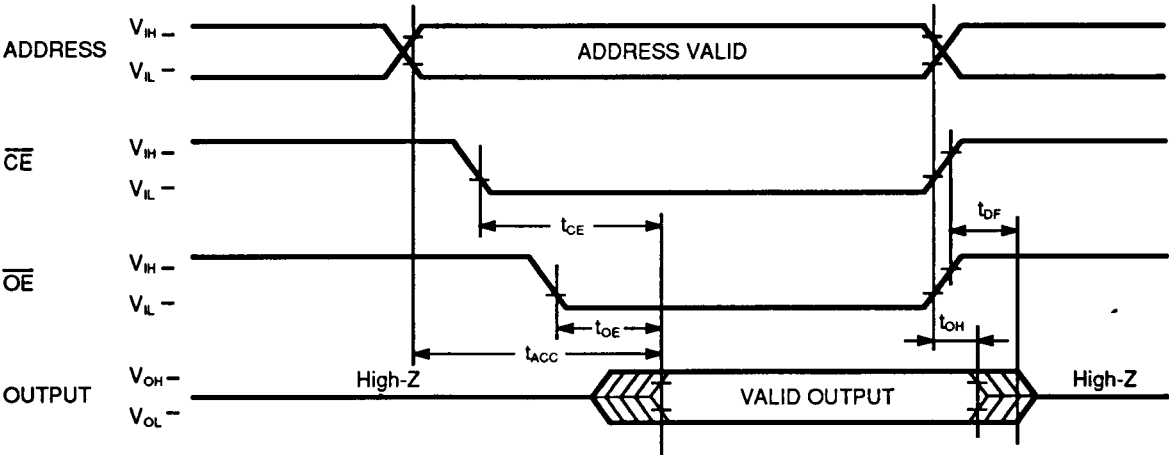
AC CHARACTERISTICS

(Recommended operating conditions unless otherwise noted)

Parameter	Symbol	MBM27C512-20-X			Unit
		Min	Typ	Max	
Address Access Time*1	t_{ACC}			200	ns
$\overline{CE}$ to Output Delay	t_{CE}			200	ns
$\overline{OE}$ to Output Delay*1	t_{OE}			70	ns
Address to Output Hold	t_{OH}	0			ns
Output Enable High to Output Float*2	t_{DF}	0		60	ns

Notes: *1 $\overline{OE}$ may be delayed up to $t_{ACC}-t_{OE}$ after the falling edge of $\overline{CE}$ without impact on t_{ACC} .
*2 t_{DF} is specified from $\overline{OE}$ or $\overline{CE}$, whichever occurs first. Output Float is defined as the point where data is no longer driven.

OPERATION TIMING DIAGRAM



PROGRAMMING/ERASING INFORMATION

PROGRAMMING

Upon delivery from Fujitsu, or after each erasure (see Erasure section), the MBM27C512 has all 524,288 bits in the "1", or high state. "0's" are loaded into the MBM27C512 through the procedure of programming.

The MBM27C512 is programmed with a fast programming algorithm designed by Fujitsu called Quick Pro™. The programming mode is entered when +12.5V and +6V are applied to V_{PP} and V_{CC} respectively, and $\overline{CE}$ is V_{IH} . A 0.1 μ F capacitor between V_{PP} and GND is needed to prevent excessive voltage transients which could damage the device. The address to be programmed is applied to the proper address pins. The 8 bit data pattern to be written is placed on the respective data output pins. The voltage levels should be standard TTL levels. When both the address and data are stable, a 1 ms programming pulse is applied to $\overline{CE}$ and after

that one additional pulse which is 3 times as wide as previous pulse is applied to $\overline{CE}$ to accomplish the programming.

Procedure of Quick Programming (Refer to the attached flowchart.)

- 1) Set the start address (=G) at the address pins.
- 2) Set $V_{CC} = 6V$, $V_{PP} = 12.5V$ and $\overline{CE} = V_{IH}$.
- 3) Clear the programming pulse counter ($X \leftarrow 0$).
- 4) Input data to respective pins.
- 5) Apply ONE Programming pulse ($t_{PW} = 1ms$ Typ.) to $\overline{CE}$.
- 6) Increment the counter ($X \leftarrow X+1$).
- 7) Compare the number (=X) of applied programming pulse with 25 and then verify the programmed data. If programmed data is verified, go to the next step regardless of X value. If X = 25 and programmed data is not verified, the device fails. If $X < 25$ and programmed data is not verified, go back to the step

5).

- 8) Apply one additional wide programming pulse to $\overline{CE}$ (3X ms).
- 9) Compare the address with an end address (=N). If the programmed address is the end address, proceed to the next step. If not, increment the address ($G \leftarrow G+1$) and then go to the step 3) for the next address.
- 10) Set $V_{CC} = V_{PP} = 5V$.
- 11) Verify the all programmed data. If the verification succeeds, the programming completes. If any programmed data is not the same as original data, the device fails.

A continuous TTL low level should not apply to $\overline{CE}$ input pin during the program mode ($V_{PP} = 12.5V$ and $V_{CC} = 6V$) because it is required that one programming pulse width does not exceed 78.75 ms at each address.

ERASURE

In order to clear all locations of their programmed contents, it is necessary to expose the MBM27C512-W to an ultraviolet light source. A dosage of 15 W-seconds/cm² is required to completely erase an MBM27C512. This dosage can be obtained by exposure to an ultraviolet lamp (wavelength of 2537 Angstroms (Å)) with intensity of 12000 μ W/cm² for 15 to 21

minutes. The MBM27C512 should be about one inch from the source and all filters should be removed from the UV light source prior to erasure.

It is important to note that the MBM27C512 and similar devices, will erase with light sources having wavelengths shorter than 4000 Å. Although erasure time will be much longer than with UV

source at 2537 Å, nevertheless the exposure to fluorescent light and sunlight will eventually erase the MBM27C512, and exposure to them should be prevented to realize maximum system reliability. If used in such an environment, the package windows should be covered by an opaque label or substance.

ELECTRONIC SIGNATURE

The MBM27C512 has electronic signature mode which is intended for use by programming equipment for the purpose of automatically matching the device to be programmed with its corresponding pro-

gramming algorithm.

The electronic signature is activated when +12V is applied to address line A_9 (pin 24) of the MBM27C512. Two identifier bytes

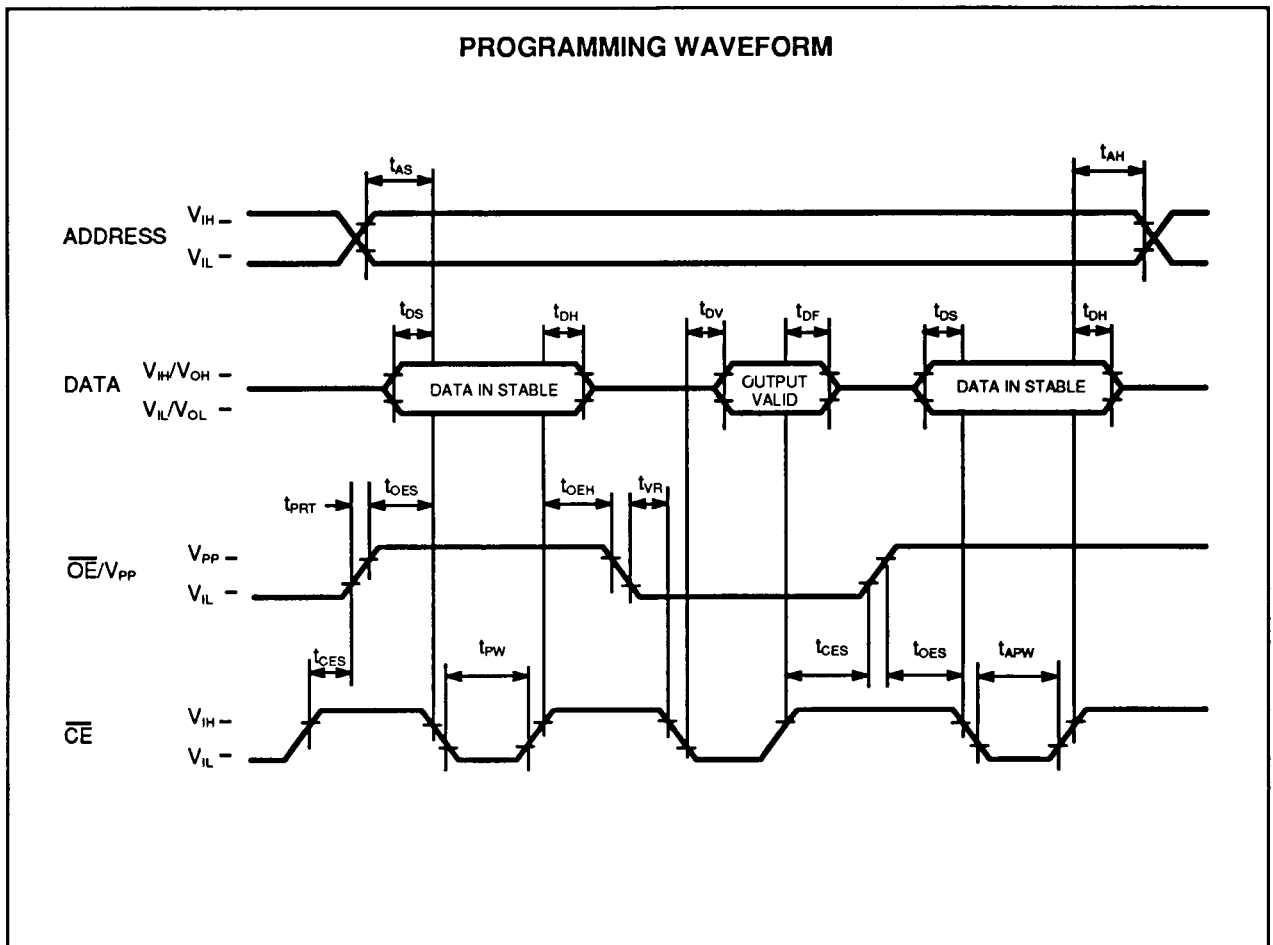
are read out from the outputs by toggling address line A_0 (pin 10) from V_{IL} to V_{IH} . The address lines from A_1 to A_{13} must be hold at V_{IL} to keep the electronic signature mode. See the table below.

A_0	O_1	O_2	O_3	O_4	O_5	O_6	O_7	O_8	Definition
V_{IL}	0	0	1	0	0	0	0	0	Manufacture
V_{IH}	1	1	0	0	0	1	1	1	Device

Note: $A_9 = 12V \pm 0.5V$

A_1 thru $A_8 = A_{10}$ thru $A_{13} = \overline{CE} = \overline{OE} = V_{IL}$.

$A_{14} = A_{15} = \text{Either } V_{IL} \text{ or } V_{IH}.$

PROGRAMMING/ERASING INFORMATION (Continued)

DC CHARACTERISTICS(T_A = 25±5°C, V_{CC}*1 = 6V±0.25V, V_{PP}*2 = 12.5V±0.3V)

Parameter	Symbol	Value			Unit
		Min	Typ	Max	
Input Leakage Current (V _{IN} = 6.25V/0.45V)	I _U			10	μA
V _{PP} Supply Current During Programming Pulse ($\overline{CE}$ = V _{IL})	I _{PP}			50	mA
V _{CC} Supply Current	I _{CC}			30	mA
Input Low Level	V _{IL}	-0.1		0.8	V
Input High Level	V _{IH}	2.0		V _{CC} + 0.3	V
Output Low Voltage During Verify (I _{OL} = 2.1 mA)	V _{OL}			0.45	V
Output High Voltage During Verify (I _{OH} = -400μA)	V _{OH}	2.4			V

Note: *1 V_{CC} must be applied either coincidently or before V_{PP} and removed either coincidently or after V_{PP}.

*2 V_{PP} must not be 13 volts or more including overshoot. Permanent device damage may occur if the device is taken out or put into socket remaining V_{PP} = 12.5 volts. Also, during $\overline{CE}$ = V_{IL}, V_{PP} must not be switched from 5 to 12.5 volts or vice-versa.

PROGRAMMING/ERASING INFORMATION (Continued)

AC CHARACTERISTICS

($T_A = 25 \pm 5^\circ\text{C}$, $V_{CC} = 6V \pm 25V$, $V_{PP} = 12.5V \pm 0.3V$)

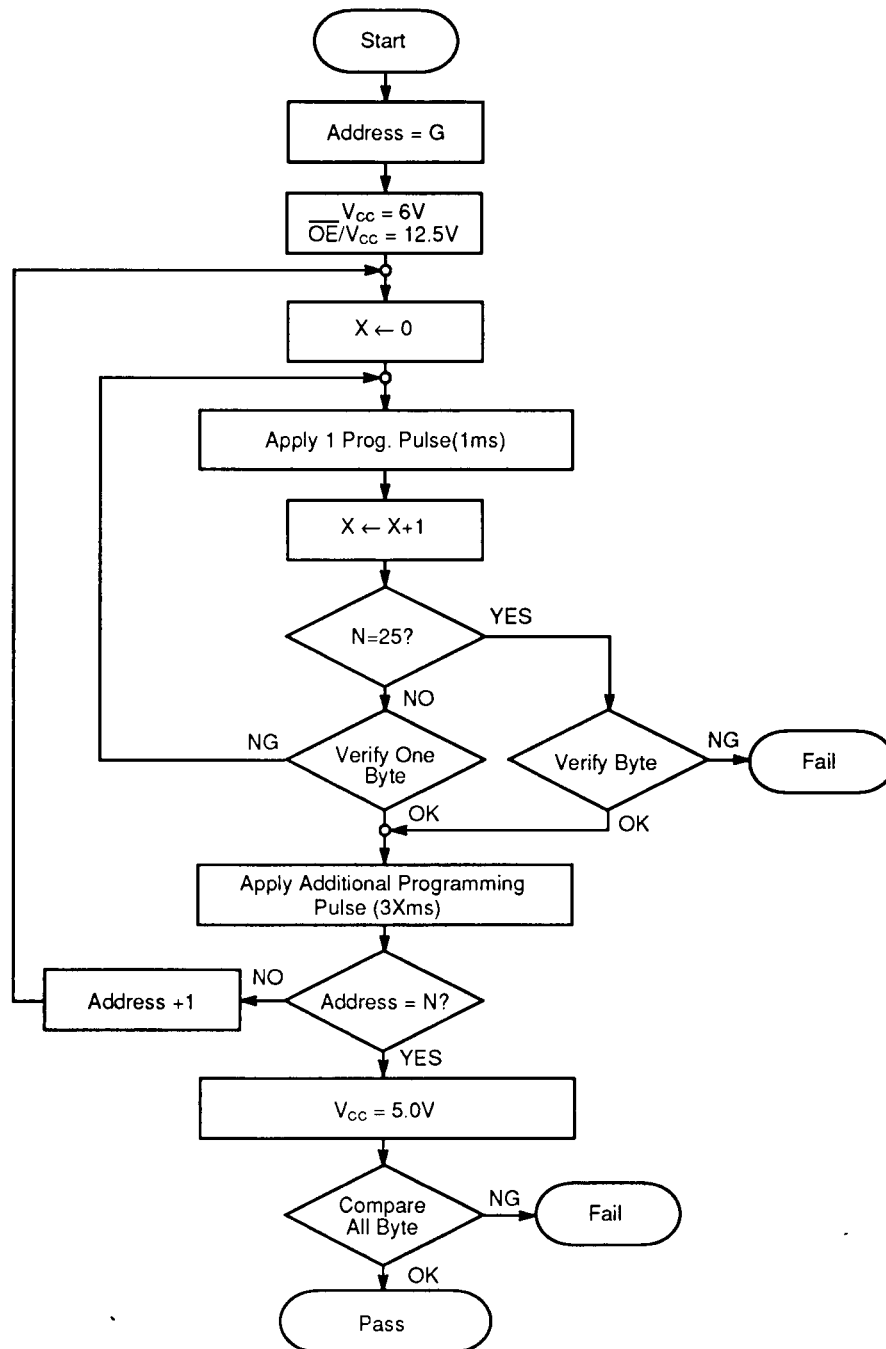
Parameter	Symbol	Value			Unit
		Min	Typ	Max	
Address Setup Time	tAS	2			μs
Chip Enable Setup Time	tCES	2			μs
Output Enable Setup Time	tOES	2			μs
Data Setup Time	tDS	2			μs
VCC Setup Time	tVS	2			μs
Address Hold Time	tAH	2			μs
Data Hold Time	tDH	2			μs
Output Enable Hold Time	tOEH	2			μs
VPP Recovery Time	tVR	2			μs
Chip Enable to Data Valid	tDV			1	μs
Output Disable to Output Float Delay	tDF	0		130	ns
VPP Program Pulse Rise Time	tPRT	50			ns
Programming Pulse Width	tPW	0.95	1	1.05	ms
Additional Programming Pulse Width	tAPW	2.85		78.75	ms

PROGRAMMING FLOWCHART

$V_{CC} = 6V \pm 0.25V$
 $\overline{OE}/V_{PP} = 12.5 \pm 0.3V$

G : Start Address
 N : Stop Address
 X : Counter Value

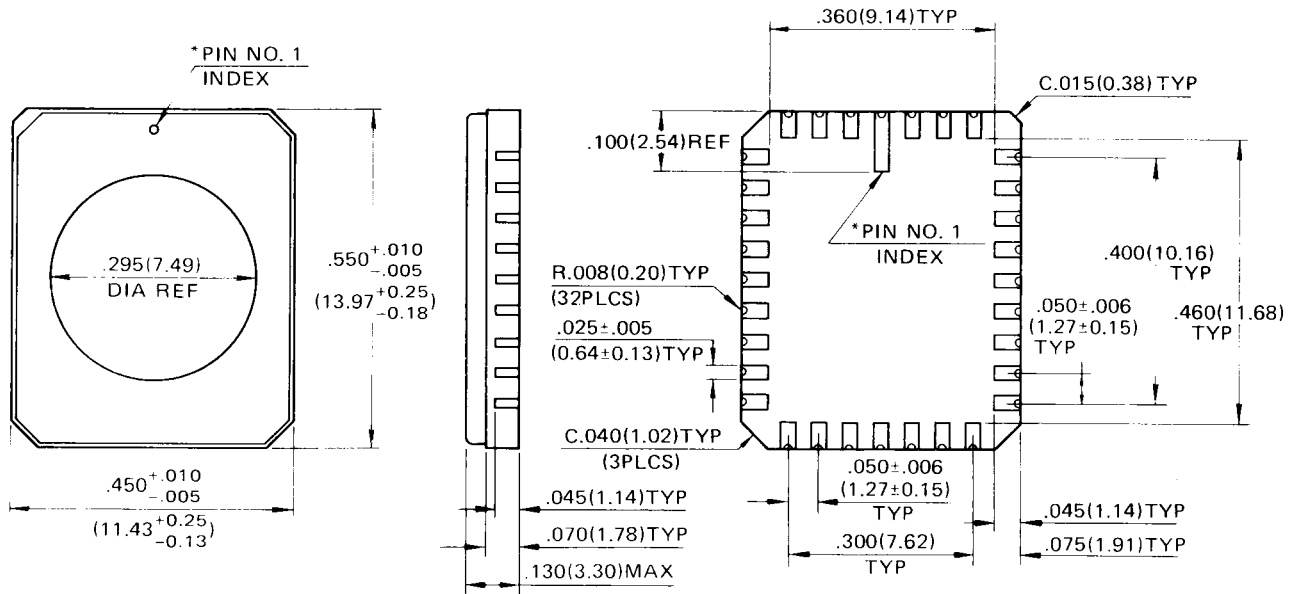
Maximum 105ms/Byte
 Minimum 3.8ms/Byte



PACKAGE DIMENSIONS

Standard 32-pad Ceramic LCC (Suffix: TV)

32-PAD CERAMIC (FRIT SEAL) LEADLESS CHIP CARRIER (CASE No.: LCC-32C-F01)



*Shape of PIN NO.1 INDEX: Subject to change without notice.

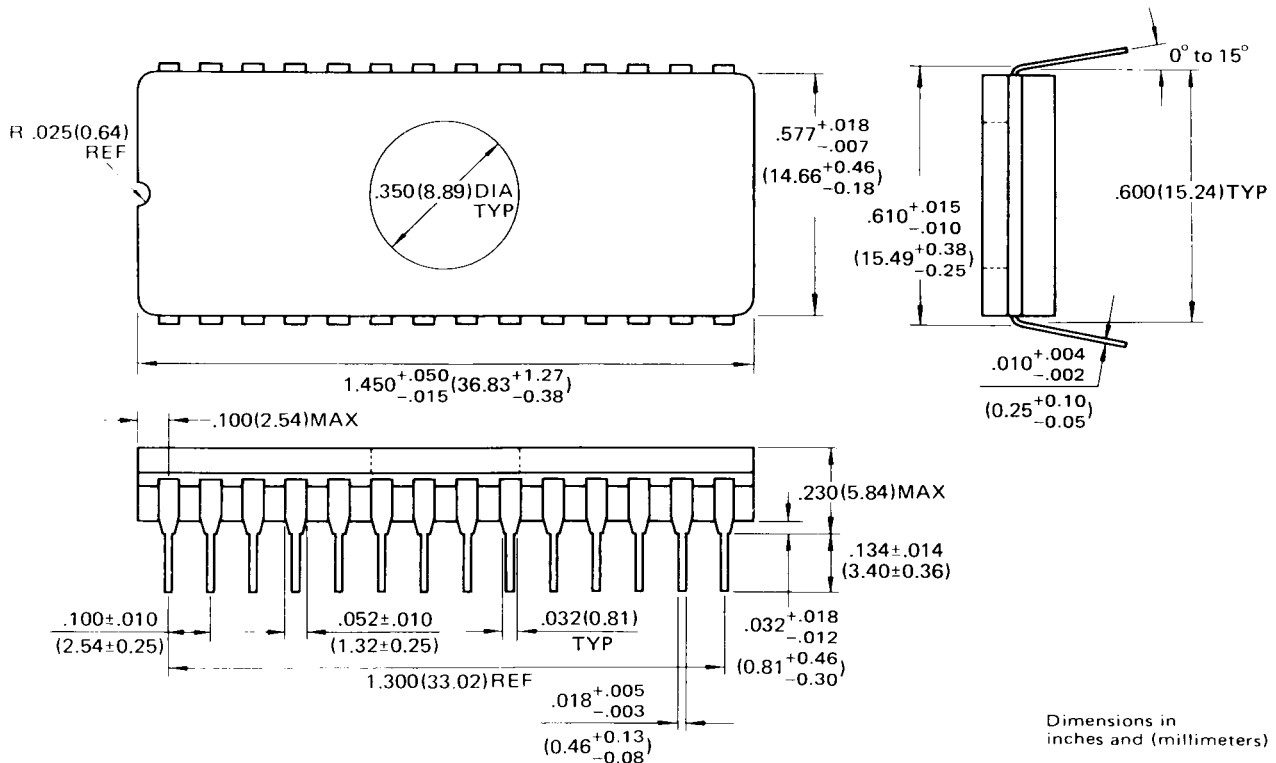
© 1988 FUJITSU LIMITED C32017S-2C

Dimension in inches
and (millimeters)

PACKAGE DIMENSIONS (Continued)

Standard 28-pin Ceramic DIP (Suffix: Z)

28-LEAD CERAMIC (CERDIP WITH TRANSPARENT LID) DUAL IN-LINE PACKAGE (CASE No.: DIP-28C-C01)



© 1988 FUJITSU LIMITED D28007S-3C

All Rights Reserved.

Circuit diagrams utilizing Fujitsu products are included as a means of illustrating typical semiconductor applications. Complete Information sufficient for construction purposes is not necessarily given.

The Information contained in this document has been carefully checked and is believed to be reliable. However, Fujitsu assumes no responsibility for inaccuracies.

The Information contained in this document does not convey any license under the copyrights, patent rights or trademarks claimed and owned by Fujitsu.

Fujitsu reserves the right to change products or specifications without notice.

No part of this publication may be copied or reproduced in any form or by any means, or transferred to any third party without prior written consent of Fujitsu.